HEIRSTONE CONSULTING

AIRPORTS AND TERMINALS

The busiest hour

How airports work, how to compare them fairly across regions and what makes a good one, with a view from the Mediterranean, Africa and Central Asia



THE REPORT IN SIX NUMBERS

1.83x

Antalya's busiest month against its average month in 2025. Run for a full year, that month would make a 71-million-passenger airport; the airport carried 39 million.

40 to 209

passengers per aircraft movement in 2025, from Zanzibar to Dubai: the same runway can serve very different airports.

9.7 to 16.9

square metres of terminal per 1,000 annual passengers of capacity at most new terminals, while cost per passenger of capacity varies more than tenfold.

30.94M

passengers at Cairo in 2025, up 6.8%, the most of any African airport on FRACS data. Egypt's public airports carried about 58.9 million.

49%

of world passenger traffic passes through airports with private investment, on ACI's 2024 ownership data; in Africa the share is 24.3%.

65

airports in the Heirstone Airport Benchmark 2026, 49 of them in the Mediterranean, Africa, Central Asia and the Caucasus.

CONTENTS

	What we found	3
1	Built for one hour	4
2	Paid for by the average passenger	9
3	Comparing airports fairly	11
	The Heirstone Airport Benchmark 2026	15
4	Good at its job	17
5	Who owns the airport	20
6	From Casablanca to Almaty	23
7	What to decide, and when	28
	Notes, sources and definitions	30

ABOUT THIS REPORT

This report explains how an airport works as a system and as a business, sets out a fair way to compare airports across very different markets, and proposes a test of what makes a good one. It is written for ministries of transport, civil aviation and tourism, airport owners and concession grantors, development finance institutions, investors and operators, and developers of airport-city land.

It draws on public sources current in 2026: airport operators, regulators, ACI World, IATA, development finance institutions, listed-company filings and national statistics. 2025, the latest full year, is the reference year. Where we combine published figures to produce a new one, such as passengers per movement, utilisation or cost per passenger of capacity, the calculation is labelled Heirstone analysis.

THE ARGUMENT

An airport is sized for its busiest hour and earns from its average passenger, and whether it is good depends on the job it does. Annual traffic, the usual yardstick, shows none of this.

WHAT WE FOUND

An airport is built for its busiest hour and paid for by its average passenger. Annual traffic, the figure every ranking leads with, measures neither.

We assembled 65 airports across the Mediterranean, Africa, Central Asia, the Gulf, Europe and Asia into one benchmark and compared them by peak, aircraft size, capacity, space, cost and traffic mix. On these measures the league table reorders, and the answer to what makes a good airport depends on the job the airport does.

1 **The peak sets the size of the airport.**

Antalya carried 39.0 million passengers in 2025. Its busiest month, run for a year, would be 71.5 million, close to the 80 million its new concession is building towards. Istanbul's busiest month was only 1.18 times its average. Leisure gateways should be planned and judged on peak-month capacity and off-peak yield.

2 **The same runway serves very different airports.**

Passengers per aircraft movement ran from 40 at Zanzibar to 209 at Dubai in 2025. Addis Ababa handled almost as many movements as Berlin with half the passengers. Gatwick moved 259,000 flights on one runway, more than Doha's 141,000 per runway.

3 **Space converges; cost does not.**

Most new terminals sit between 9.7 and 16.9 m^2 per 1,000 annual passengers of capacity. Cost per passenger of capacity runs from about US$20 for brownfield expansions to US$240 to 266 at Beijing Daxing, Rwanda's Bugesera and Heathrow's Terminal 5, the first two on press figures. Scope explains most of the spread. Cairo's Terminal 4, at US$87.5 to 112.5 per added passenger on official figures, sits within the range for new terminals.

4 **A good airport is one that fits its job.**

We define six airport jobs and an eight-question Heirstone Airport Test whose pass marks come from the benchmark. Doha is about 75% transfer; Incheon, almost wholly international, is 11%. The two should not share a scorecard.

5 **The contract matters as much as the concrete.**

Commercial income is 37% of airport revenue worldwide, and car parking now earns more than shops. Private capital stands behind 49% of world traffic and 24.3% in Africa. Egypt's IFC-advised programme for 11 airports keeps the assets in state ownership, is examining bundles of the kind used in Greece and Tunisia, and has prequalified seven consortia for its Hurghada pilot.

WHAT IT MEANS

Governments and owners should size and phase airports on a stated design hour and commit the next phase while there is still headroom: at 10% growth a year and four years to build, that means at about 68% of capacity.

Concession grantors and DFIs should choose the bid variable and the service targets with the same care as the capital plan.

Investors and operators should underwrite the peak month and price commercial income against regional peers.

65

airports in the Heirstone Airport Benchmark 2026

6 x 8

six airport jobs, eight questions: the Heirstone Airport Test weights each question by job



CHAPTER 1

01

Built for one hour

Every part of an airport, from the runway to the kerb, is sized for a design hour: a busy hour of a busy day that the owner chooses to serve well. The annual total follows from that choice.

An airport is a chain of processors. Airside, aircraft land, taxi and park at a stand. In the terminal, passengers check in, pass security and border control, wait at the gate and collect their bags. Landside, they arrive and leave by road, rail or coach. IATA's planning guidance rates each processor on space and on waiting time, and states that the weakest sub-system sets the Level of Service of the whole terminal.

Planners do not size this chain for the year. The FAA's typical peak hour, the peak hour of the average day of the peak month, equals 0.035% of annual passengers at airports above 30 million and rises to 0.2% below 100,000. British practice uses the 30th busiest hour of the year and Zurich the 20th. Auckland designs to its 30th busiest hour, deliberately below the absolute peak.

The choice has a price. Under IATA's older A to F scale, designing for level A needed 30 to 50% more space per passenger than the middle level C. IATA now asks planners to target an Optimum band for each processor, and accepts over-design only in a start-up period of five to ten years.

The landside end of the chain grows in importance as new airports move out of town. On straight-line distances from the city centre, Incheon sits 48 km out, Istanbul 33 km and Riyadh 30 km, against under 10 km for Dubai International and Doha. Heathrow Express reaches central London in 15 minutes.

> *One under-sized processor, a security hall or a baggage reclaim, sets the service level for the whole terminal.*

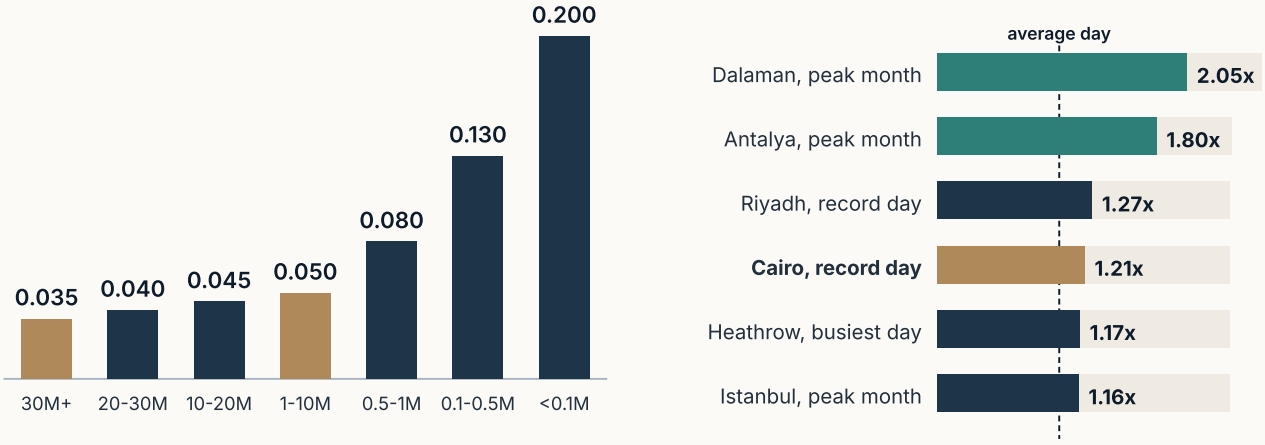
THE DESIGN HOUR

Smaller airports peak harder than large ones, and leisure airports hardest of all

EXHIBIT 1

The design hour is a small share of the year, and peaks are sharpest at small and seasonal airports

Left: FAA typical peak-hour passengers as a share of annual passengers, by annual traffic band (%). Right: busiest day, or average day of the busiest month, against the average day of 2025



Source: FAA AC 150/5360-7 ratios as reproduced in Journal of Air Transport Management, vol. 96 (2021); Cairo Ministry of Civil Aviation via Business Today Egypt (2025); Heathrow (2026); Saudi Press Agency via Asharq Al-Awsat (2025) and Bernama (2026); DHMI monthly passenger files (2025-26). Gold columns mark the bands of the two terminals discussed in the text. **Heirstone analysis** for all ratios on the right.

The ratios fall as airports grow because large airports spread their flights across the day. An airport of 30 million passengers or more plans its terminal around 0.035% of its annual traffic in the typical peak hour; one below a million plans around 0.08% or more. Operators' own figures can be checked against it. Abu Dhabi's Terminal A is stated at 11,000 passengers an hour for 45 million a year, or 0.024%, below the FAA ratio for its size. Samarkand's 2022 terminal, at 1,000 an hour for 2 million a year, matches its band exactly at 0.05%.

Abu Dhabi's figure suggests that hubs with banks of connecting flights can be sized below generic ratios; small airports cannot. Leisure airports face a second peak on top of the daily one. In 2025 Antalya's busiest month averaged 1.80 times its annual average day, and Dalaman's 2.05 times. Capital airports run closer to 1.2. Cairo set a record of about 103,000 passengers on 696 flights at the end of 2025, 1.21 times its average day, a published peak-day figure of the kind planners need. Heathrow's busiest day, over 270,000 passengers, was 1.17 times its average and Riyadh's record 1.27 times.

The practical rule is to size each processor on the airport's own ranked hours, test it against the published ratios and state the result. A design hour that cannot be traced to data will be argued over at every later stage, from the tender to the regulator's price review.

0.035%

of annual passengers in the typical peak hour, the FAA ratio for airports of 30 million or more

1.21x

Cairo's record day against its 2025 average day (Heirstone analysis)

30th

busiest hour of the year, the British design standard; Zurich uses its 20th

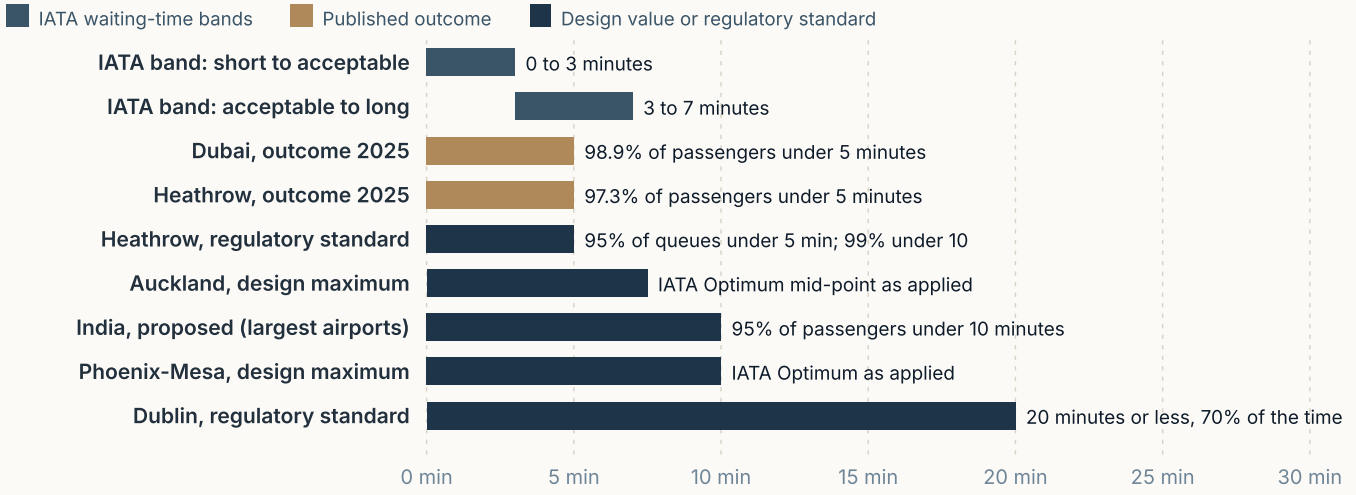
LEVEL OF SERVICE

Security queues are now published as outcomes, and the best outcomes beat the design standards

EXHIBIT 2

Security standards cluster at five to ten minutes; Dubai and Heathrow publish results inside them

Maximum security waiting time in IATA bands, applied design values, regulatory standards and published operator outcomes (minutes)



Source: IATA, Level of Service position paper and IATA in International Airport Review (2015); Auckland Airport submission to the New Zealand Commerce Commission (2024); Phoenix-Mesa Gateway Airport master plan (2019); Dubai Airports (2026); Heathrow results (2026); UK CAA, H8 initial proposals (2026); AERA consultation (2025); Irish Aviation Authority.

IATA publishes its Level of Service framework in concept, but not the numeric space table that sits behind it. What is public are its 2015 waiting-time bands: 0 to 3 minutes at security is short to acceptable and 3 to 7 minutes acceptable to long; economy check-in runs to 12 and then 30 minutes, outbound passport control to 5 and 10, and baggage reclaim to 12 and 18. Airports that apply the bands show how they read them. Auckland sizes gate lounges at 1.8 m^2 per seated and 1.2 m^2 per standing passenger, 1.5 m^2 at an even mix, and caps the security queue at 7.5 minutes.

The stronger comparator is the outcome. Dubai International reported that 98.9% of its 95.2 million passengers waited under five minutes at security in 2025, 99.35% under ten minutes at departure passport control and 98.8% under fifteen minutes on arrival, with 89% of bags delivered within 45 minutes and 2.47 mishandled bags per 1,000 passengers. Heathrow reported 97.3% under five minutes at security for 2025. Its regulator's standard is 95% of queues under five minutes and 99% under ten. India's regulator has proposed 95% under ten minutes at its largest airports. These figures can be written straight into a concession or management contract.

Capacity can be declared the same way. Gatwick's coordinator declares terminal limits in passengers an hour as well as runway slots: 5,400 an hour through the North Terminal's departure lounge and check-in, 3,800 through the South Terminal's departure lounge. Regional airports that today declare only runway slots could adopt the same template.

98.9%

of Dubai International's passengers waited under five minutes at security in 2025

1.5 m^2

per passenger in Auckland's gate lounges, the low end of IATA's Optimum band as applied

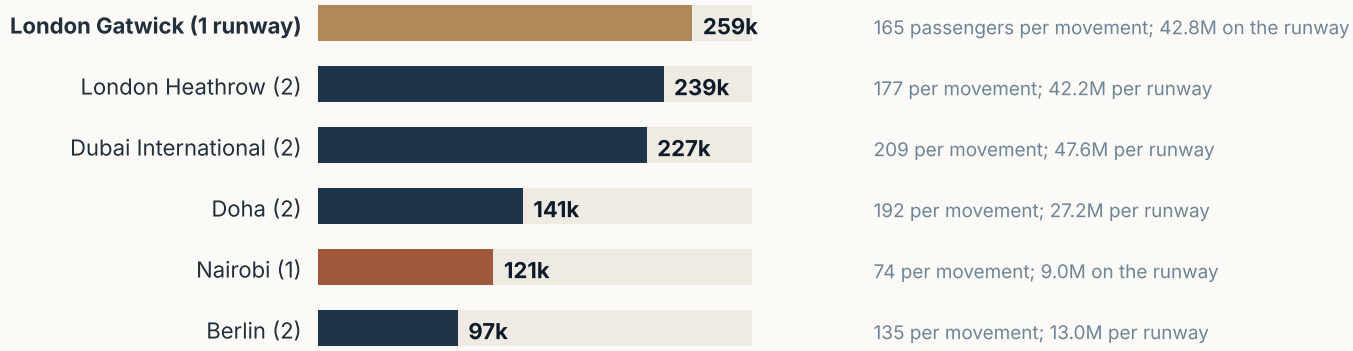
RUNWAYS AND STANDS

Aircraft size and runway discipline decide output more than the number of runways

EXHIBIT 3

Gatwick's single runway handled more movements in 2025 than each of Heathrow's two

Aircraft movements per runway in use, 2025, with passengers per movement and passengers per runway

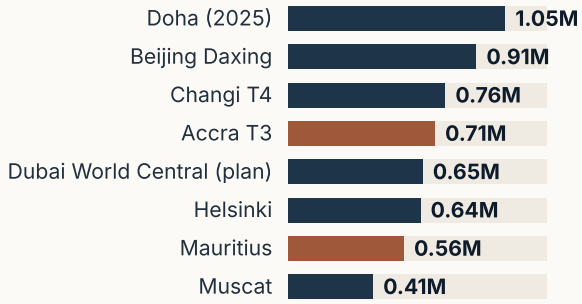


Source: Gatwick Airport Limited, financial statements 2025; Heathrow (2026); Dubai Airports (2026); HIA, Doha (2026); FRACS bulletin (2026); Flughafen Berlin Brandenburg (2026); Greater London Authority (2025). Gatwick has two runways but only one can operate at a time. **Heirstone analysis.**

EXHIBIT 4

Passengers per contact stand vary 2.5-fold

Annual passengers of design capacity per contact stand or boarding bridge (million)



Source: Operators, architects and contractors as listed in the notes; Helsinki derived from Finavia's stated increases; Muscat on the contractor's 12 million (20 million in other sources gives 0.69). **Heirstone analysis.**

Heathrow's planning conditions cap it at 480,000 air transport movements a year. It used 98.7% of the cap in 2024 and 99.6% in 2025, the clearest public case of a cap that binds. Gatwick's declared limit was 55 movements an hour in summer 2025, rising to 57 from summer 2026, and its one runway carried 42.8 million passengers in 2025. Nairobi's 2026 master plan proposes a second runway 1,525 metres from the first, allowing independent operations and 61 arrivals and 80 departures an hour.

Aircraft size changes the arithmetic. Dubai averages over 200 passengers per movement, Nairobi 74, so the same hourly slot moves almost three times as many people. A second runway is therefore as much a decision about peak waves and resilience as about annual capacity.

Stands follow the same logic. Doha lists 62 contact gates and 128 remote stands; Dubai World Central plans more than 400 contact stands for 260 million passengers. Heathrow's regulator expects 95% of passengers to be served from pier-served stands, and India's draft standard asks for boarding bridges for 90% of movements. Copying a peer's stand count without its fleet mix misleads.

TERMINAL SHAPES

Six terminal shapes, each a different trade between walking distance, stands and phasing

EXHIBIT 5

Terminal typologies with named examples

Schematic plans (not to scale); gold dots represent aircraft stands. Figures as stated by operators and architects



Compact

Changi Terminal 4 (2017)

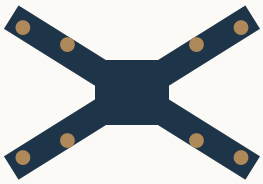
195,000 m² of floor area for 16 million passengers and 21 contact stands; facial-recognition self-service expected to save about 20% of manpower over time.



Pier extension

Doha, Munich, Incheon

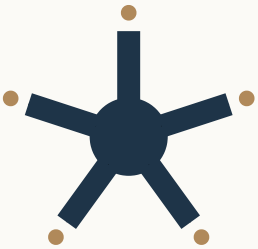
Doha's 2025 concourses added 17 contact gates; Munich's Terminal 1 pier adds 95,000 m² and 6 million passengers for a self-funded €665 million; Incheon extended Terminal 2 by 600 m at both ends.



X-shaped midfield

Abu Dhabi Terminal A (2023)

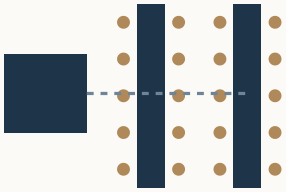
742,000 m² of built-up area for 45 million passengers; the architect states the X plan can reach 80 million.



Radial

Beijing Daxing (2019)

700,000 m² and 79 gates on airbridges, with the farthest gate under eight minutes' walk, according to its architects.



Satellite concourses

Dubai World Central (planned)

Four satellite concourses with more than 400 contact stands for 150 million passengers within ten years and 260 million ultimately.



Single roof

Istanbul (2018-19)

A 1.4 million m² terminal under one roof for 90 million passengers in the first phase and 200 million at full build-out.

Source: Changi Airport Group (2013, 2017); HIA, Doha (2025); Munich Airport (2026); Incheon International Airport Corporation and Korea Times (2024); Gulf News quoting Abu Dhabi Airports, and KPF (2023); ZHA and ADP Ingénierie (2019); Dubai Airports, Dubai World Central fact file; iGA Istanbul Airport. Shapes drawn by Heirstone.

Each shape answers the same question differently: how to put as many aircraft as possible against the building while keeping walking distances short and leaving room to grow. Piers are cheap to extend, which is why Doha, Munich and Incheon added capacity that way. Radial and X-shaped plans shorten walks from a central hall. Satellites trade an underground link for a very large number of contact stands.

IATA's guidance is that walks of more than 300 metres should be helped by moving walkways, and US planning guidance uses a similar threshold of about 1,000 feet. Doha publishes its transfer walks: one to four minutes to nearby concourses and eleven minutes to the farthest gate. Shape also fixes phasing: a terminal that cannot grow along a pier or into a satellite will need a second building and a second forecourt.

300 m

walking distance above which IATA guidance calls for moving walkways

11 min

Doha's published walk from its transfer area to its farthest gate

CHAPTER 2

02

Paid for by the average passenger

Airports earned US$194.9 billion in 2024, according to ACI World, still 2.1% below 2019 in real terms. Aeronautical charges paid by airlines supplied 54% and commercial income 37%. ACI puts the industry's return on invested capital at 6.3%, below its weighted average cost of capital.

Much of an airport's cost is fixed by the capacity it builds, and that capacity is set by the design hour. Most of its revenue arrives one passenger at a time. The gap between the two is widest at seasonal airports, which carry peak-sized assets through quiet months.

EXHIBIT 6

An average passenger brought in US$20.59 and cost US$15.64 in 2024

Airport revenue and cost per passenger, world and Asia-Pacific averages, FY2024 (US$)



Source: ACI World, Airport Economics data via ACI Insights (2026) and ACI information paper to the ICAO Asia-Pacific DGCA conference (2026). Other revenue is the residual of the published total. The Asia-Pacific cost split is not published.

The world averages hide a wide regional spread. Asia-Pacific airports earned US$14.03 per passenger in 2024 against the world's US$20.59, with lower charges and lower costs. Middle East airports carried the highest unit costs in ACI's 2022 data, US$20.82 of operating and US$9.76 of capital cost per passenger, against US$8.42 and US$4.25 in Africa. Their capital spending per passenger in 2015 to 2019 averaged US$17.83, almost three times Africa's US$6.30.

Regulation can push yields far above the average. Heathrow's 2026 charges are set to recover a maximum of £26.22 per passenger, about US$34.7, roughly three times the world's aeronautical revenue per passenger. Its departing charge for an origin passenger flying beyond Europe is £49.10, against £29.46 for a transfer passenger on the same routes. ACI estimates capacity investment needs of more than US$4 trillion to 2055.

6.3%

return on invested capital for the world's airports in 2024, below their cost of capital (ACI)

£26.22

maximum yield per passenger in Heathrow's 2026 charges, about US$34.7

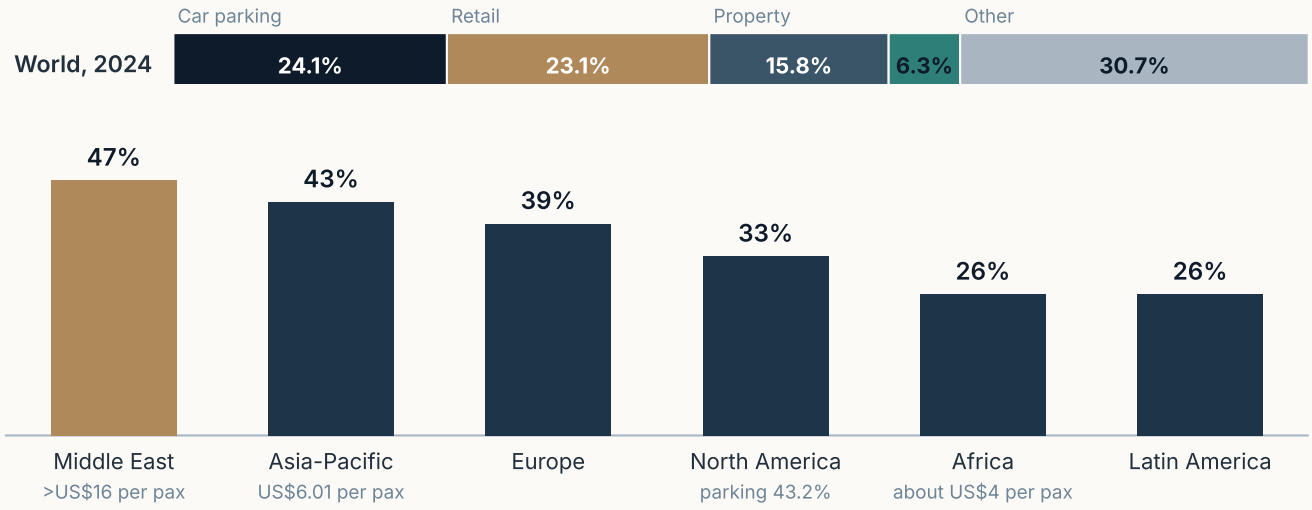
COMMERCIAL INCOME

Car parking now earns airports more than shops do, and the regional gap in commercial income is fourfold

EXHIBIT 7

Parking is the largest commercial line worldwide; the Middle East earns the highest share, Africa and Latin America the lowest

Top: non-aeronautical revenue by category, world, FY2024 (%). Bottom: non-aeronautical share of airport revenue by region, latest ACI data (%), with revenue per passenger where published



Source: ACI World Insights (2025, 2026); ACI information paper to ICAO (2026); TRBusiness reporting ACI data presented at MEADFA (2025) for the Middle East and Africa per passenger figures (2023). Other includes advertising, car rental and other concessions. Regional shares are ACI's latest year in a 2025 publication.

Non-aeronautical revenue per passenger slipped from US$8.61 to US$7.57 by 2024 on ACI's figures. Car parking, at 24.1% of the total, has overtaken retail at 23.1%, with property at 15.8% and food and beverage at 6.3%. In North America parking is 43.2%; in Asia-Pacific and the Middle East retail leads at 38.8%. Middle East airports earned over US$16 of commercial revenue per passenger in 2023, with retail at 48% of it, and African airports about US$4.

Definitions drive the headline numbers. Dubai Duty Free sold US$2.378 billion in 2025, about US$25 per Dubai passenger, but those are retailer sales. Paris's Extime retail sales were €31.7 per passenger. Heathrow's own retail revenue, which includes parking and catering, was about £9.36 per passenger, and Frankfurt's net retail revenue €3.24 in the first half of 2026. A benchmark must convert each to one definition before comparing.

For leisure gateways the lesson is direct. Where most passengers arrive by coach or hotel transfer, parking-led commercial plans imported from capital airports will not hold. Retail, food and beverage, property and the design of charges carry the commercial case.

24.1%

of non-aeronautical revenue comes from car parking worldwide, ahead of retail at 23.1%

US$25

Dubai Duty Free sales per Dubai International passenger, 2025 (retailer sales, Heirstone analysis)

48%

of Middle East airports' commercial revenue came from retail in 2023

CHAPTER 3

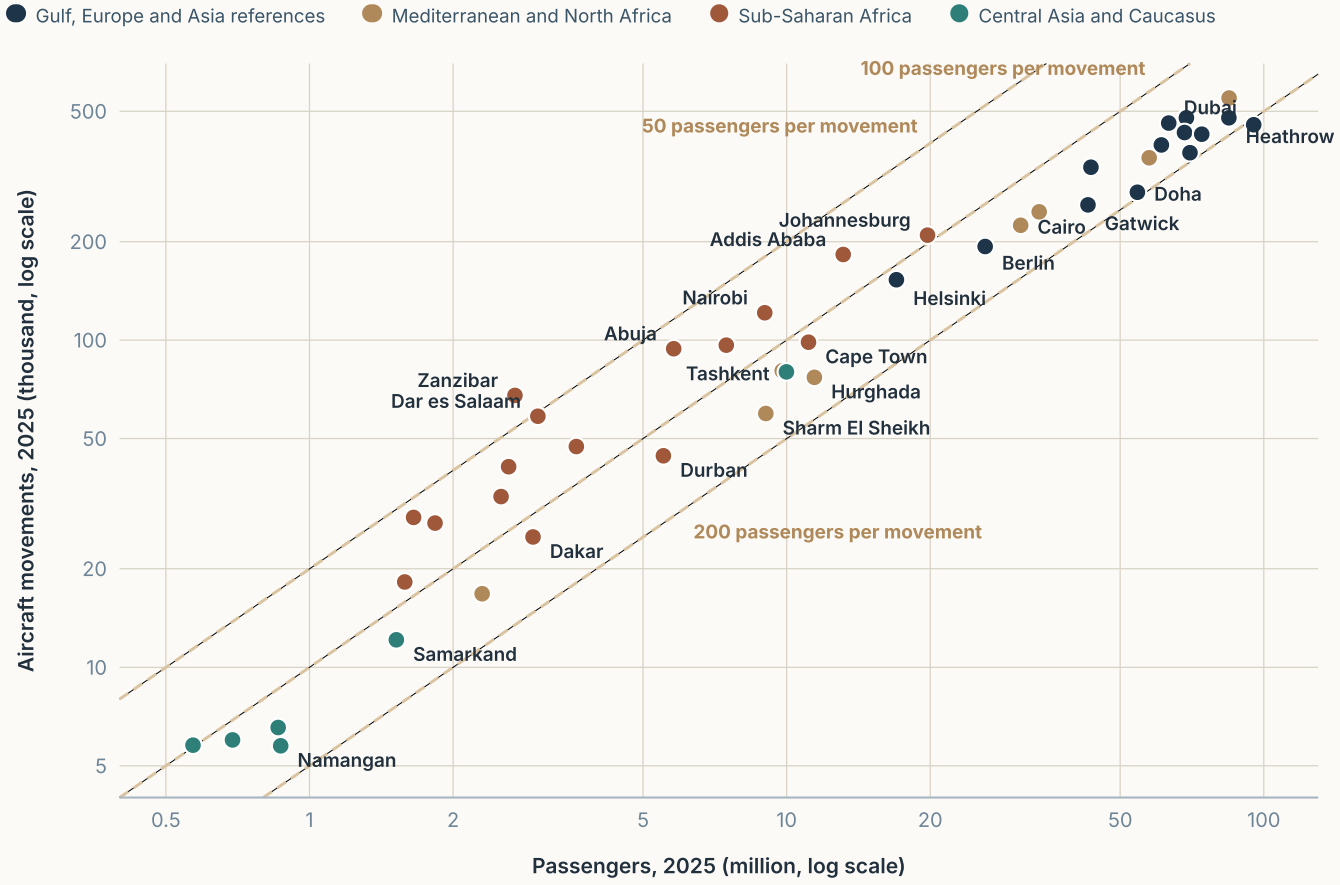
03 Comparing airports fairly

Annual passengers rank airports by size. They say little about how hard an airport works or how well it fits its job. We normalise in six ways: by aircraft size, by peak, by stated capacity, by terminal space, by cost and by traffic mix. Each changes the order.

EXHIBIT 8

Addis Ababa handles almost as many flights as Berlin with half the passengers

Passengers and aircraft movements, 2025, benchmark airports with both figures; dashed lines show equal passengers per movement



Source: Heirstone Airport Benchmark 2026: operators, DHMI, Aena, FRACS (Africa), Uzbekistan Airports via Anhor.uz and UzDaily, and others listed in the notes. Movements include all aircraft movements as published. **Heirstone analysis.**

Addis Ababa handled 182,630 movements in 2025 for 13.1 million passengers, about 72 per movement; Berlin handled 193,042 for 26.1 million, 135 per movement. On the runway the two airports are alike; in the terminal Berlin is twice as busy. Zanzibar's 67,735 movements carried 2.69 million passengers, 40 per movement, while Dakar carried 2.94 million on 25,020, about 117 each. Dubai sits at the other end at 209 on a simple division, or 214 on Dubai Airports' own measure.

The spread follows the airport's job. Gulf transfer hubs with widebody fleets cluster near 200 passengers per movement; East African airports, with more regional and domestic flying, sit below 80. Runway, apron and charges planning should therefore be set per movement and per aircraft code, with passenger totals used for the terminal.

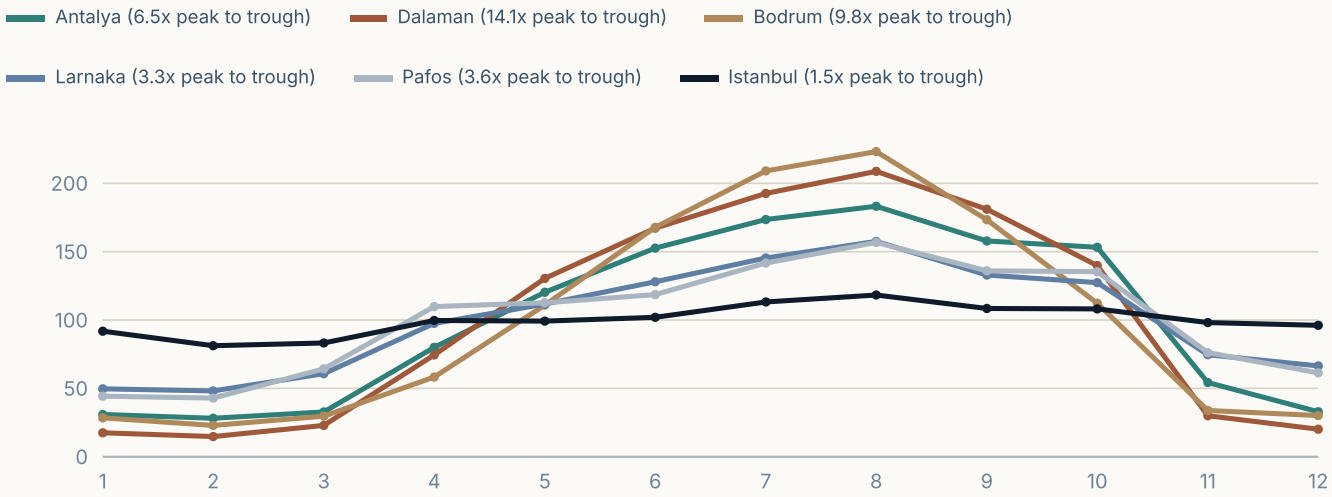
NORMALISING FOR THE PEAK

At its summer peak, a 39-million-passenger airport runs like one of 71 million

EXHIBIT 9

Leisure airports move half their year in four summer months; Istanbul barely varies

Monthly passengers in 2025 as an index of each airport's average month (average month = 100); months numbered from the start of the year

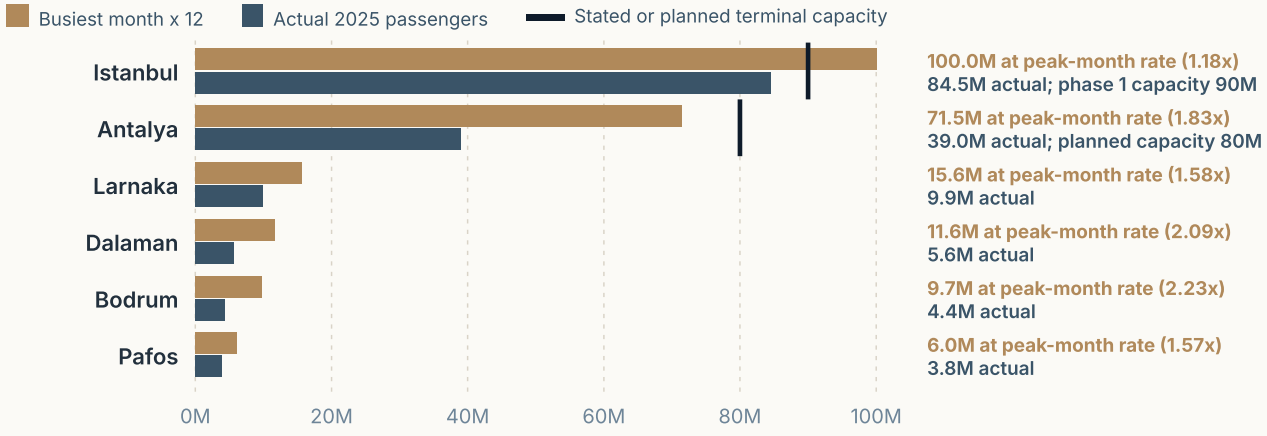


Source: DHMI monthly cumulative passenger files (months derived by differencing); Hermes Airports passenger statistics. **Heirstone analysis.**

EXHIBIT 10

Run at its busiest month for a year, Antalya would need 71 million passengers of capacity; its new concession plans for 80 million

Actual 2025 passengers against the busiest month multiplied by twelve (million), with stated or planned capacity where published



Source: DHMI; Hermes Airports; TAV Airports (2021) for Antalya's 80 million; iGA for Istanbul's 90 million phase-one capacity. **Heirstone analysis.**

The four summer months carried 55.6% of Antalya's 2025 passengers, 62.5% of Dalaman's and 64.5% of Bodrum's, against 36.8% at Istanbul. On annual figures Antalya's planned 80 million looks twice what it needs; against its peak month the plan uses 89%.

Egypt's Red Sea gateways appear to follow a flatter calendar. Hurghada handled 2.64 million passengers in the three summer months of 2025, according to the Civil Aviation Ministry as reported in the trade press, about 23% of its 11.4 million for the year on FRACS figures; Antalya's three summer months carried 42%. The two figures come from different sources, so the comparison is indicative. Monthly airport-level data would let investors price that flatter profile.

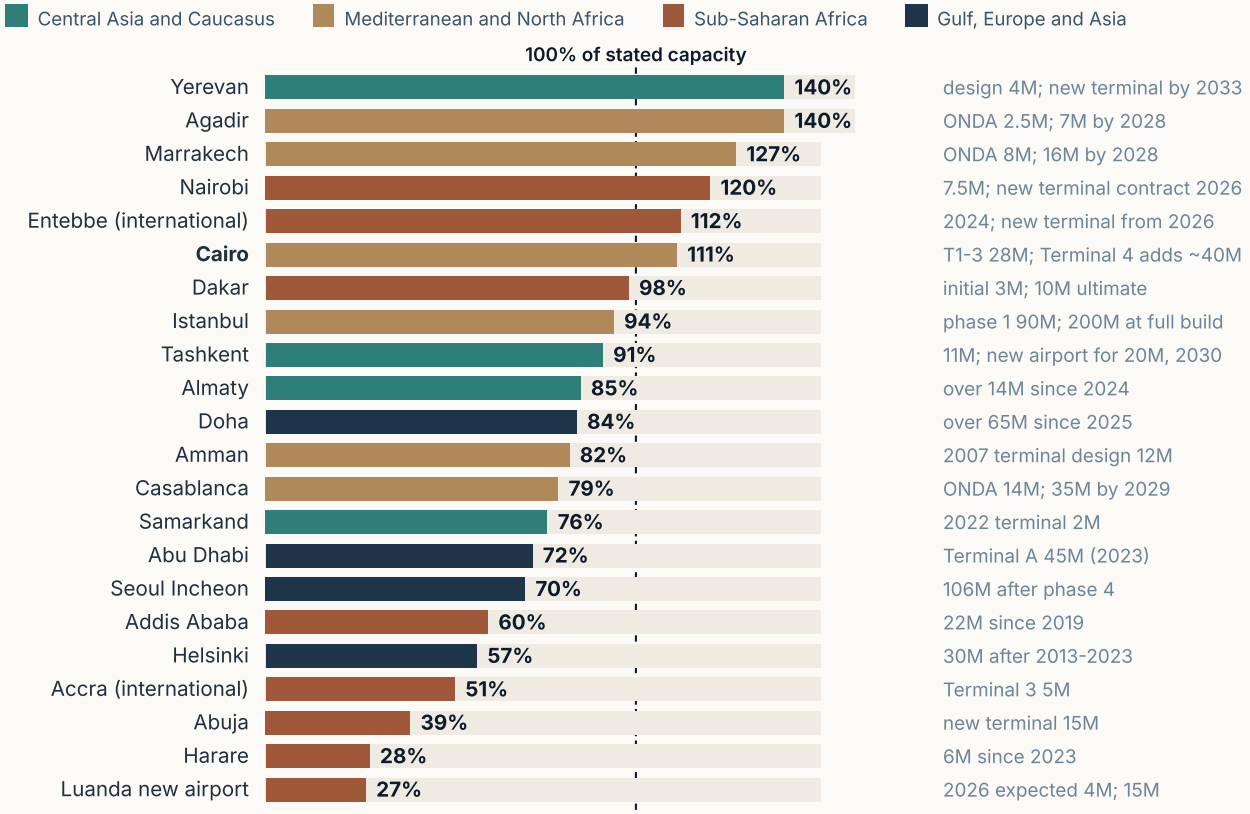
NORMALISING FOR CAPACITY

Airports in the benchmark run at 27% to 140% of stated capacity; each one above 100% has its next phase planned or under way

EXHIBIT 11

Traffic against stated capacity ranges more than fivefold

Passengers in the latest year as a share of stated or design capacity (%), with the capacity basis and the next phase where published



Source: Heirstone Airport Benchmark 2026. Capacity as stated by operators, ministries and ONDA; definitions differ (terminal design, airport-wide, initial phase). 2025 traffic except Entebbe (2024) and Luanda (traffic expected by the operator for 2026). Entebbe and Accra use international traffic and international capacity. **Heirstone analysis.**

Six airports in the sample carried more passengers in the latest year than their stated capacity, and each has its next phase planned or under way: Yerevan, with a concession extended to 2067 against about US$425 million of investment; Agadir and Marrakech, in Morocco's programme to 2028; Nairobi, where a new terminal contract was signed in 2026; Entebbe, whose new terminal opens in phases from 2026; and Cairo, where Terminal 4 adds about 40 million passengers. In each, demand has run ahead of a programme that is now moving.

At the other end, Harare at 28%, Abuja at 39% and Luanda's new airport, expected at 27% in 2026, opened capacity ahead of the airline networks that would fill it.

Headroom is a timing question. Cairo's Terminal 4, Casablanca's new terminal and the new Tashkent airport are all four-to-five-year programmes. With four years to build, the next phase should be committed at about 82% of capacity if traffic grows 5% a year, 68% at 10% and 57% at 15%.

75%

of Doha's 54.3 million passengers were transfers in 2025 (Heirstone analysis); Amsterdam 36.6%, Helsinki 32.6%, Baku about 12.7%, Incheon 10.9%

68%

of capacity: the point to commit the next phase at 10% growth and a four-year build (Heirstone analysis)

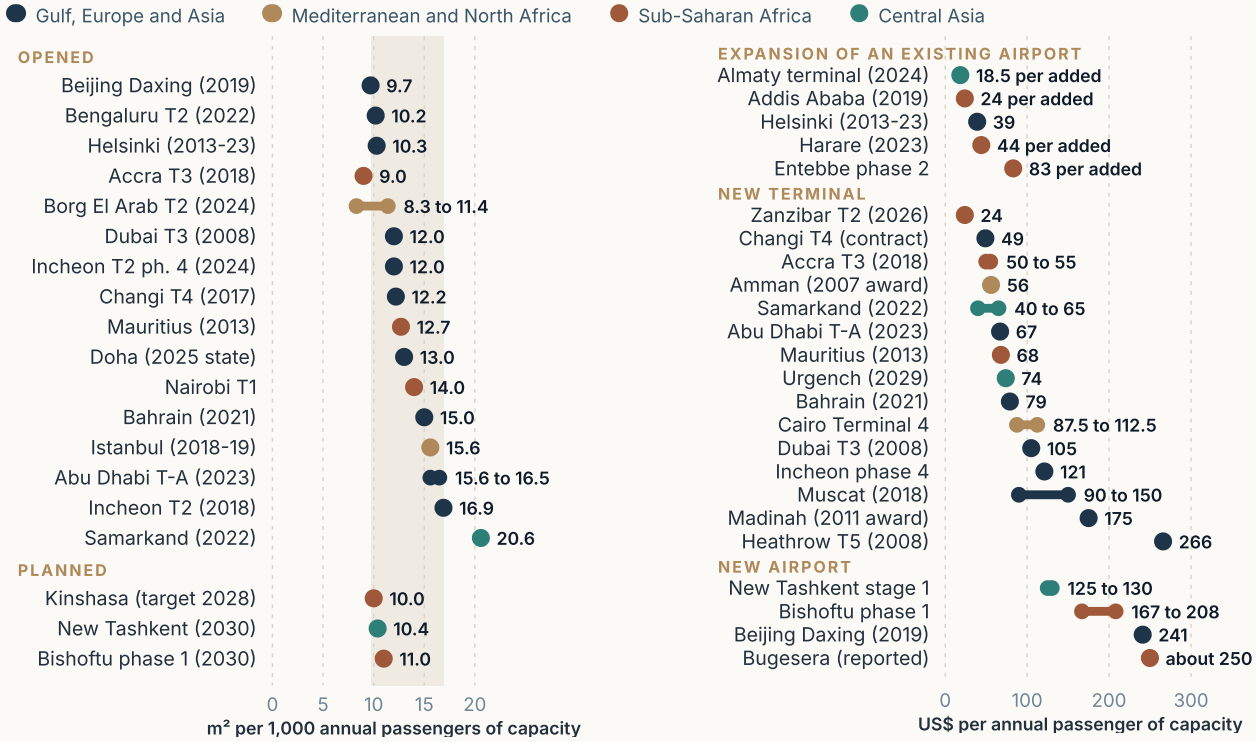
NORMALISING FOR SPACE AND COST

Space per passenger converges across regions, while cost per passenger varies more than tenfold

EXHIBIT 12

Most terminals fall in a 9.7 to 16.9 m² band; costs run from under US$20 to over US$260 per passenger of capacity

Left: gross terminal area per 1,000 annual passengers of design capacity (m²). Right: capital cost per annual passenger of capacity (US$), grouped by scope; ranges where sources differ



Source: Operators, architects, contractors, ministries and press as listed in the notes. Costs are as published and not like for like: some are construction contracts, some whole programmes including runways. Changi T4 is the main construction contract only. Daxing, Bugesera and Abu Dhabi Terminal A costs are press figures; Samarkand and Kinshasa areas are trade press. Expansion costs are per added passenger. **Heirstone analysis.**

Space is the most consistent ratio in airport planning. Eleven terminals opened since 2008 in Asia, Europe and the Gulf, with areas stated by their operators or architects, sit between 9.7 and 16.9 m² per 1,000 annual passengers of capacity. Mauritius and Nairobi's Terminal 1 sit inside the band and Accra just below it, while the band also holds the terminals planned in Ethiopia, Uzbekistan and the Democratic Republic of Congo. Anything outside the band needs a stated reason, such as a whole-campus area or car parks inside the figure.

Cost is far less consistent, and scope explains most of the spread. Brownfield expansions are cheapest: Almaty's 2024 terminal added about 10.8 million passengers of capacity for about US$200 million. Whole new airports, with runways and roads, cost US$125 to 250 per passenger. Cairo's Terminal 4, on official estimates of US$3.5 billion to US$4.5 billion for about 40 million passengers, comes to US$87.5 to 112.5, within the range of new terminals at existing airports and below every greenfield airport in the sample.

390,000 to 680,000

m² of terminal for a 40 million passenger airport on the benchmark band

US$87.5 to 112.5

Cairo Terminal 4 cost per passenger of added capacity on official figures (Heirstone analysis)

THE HEIRSTONE AIRPORT BENCHMARK 2026

Sixty-five airports, normalised

2025, the latest full year, unless noted. Blank cells were not published or could not be verified, and are left blank. Shaded columns and derived columns (passengers per movement, utilisation, cargo per 1,000 passengers) are Heirstone analysis. Job is the Heirstone classification set out in Chapter 4.

EXHIBIT 13

The Heirstone Airport Benchmark 2026 (part 1 of 2)

AIRPORT	IATA	COUNTRY	JOB	PAX 2025 (M)	GROWTH (%)	MOVEMENTS (000)	PAX PER MOVEMENT	INTL SHARE (%)	TRANSFER SHARE (%)	CAPACITY (M)	UTILISATION (%)	CARGO T PER 1,000 PAX
GULF, EUROPE AND ASIA REFERENCES												
Dubai	DXB	UAE	Global hub	95.2	+3.1	455	209	100				
Doha	DOH	Qatar	Global hub	54.3	+3.0	283	192		75	65	84	47.7
Abu Dhabi	AUH	UAE	Global hub	32.5	+13.0					45	72	
Jeddah[c]	JED	Saudi Arabia	O&D gateway	53.5	+9.0						107	
London Heathrow[a]	LHR	UK	Global hub	84.5	+0.7	478	177	94			99.6	
London Gatwick	LGW	UK	O&D gateway	42.8		259	165					
Amsterdam	AMS	Netherlands	Global hub	68.8	+2.9	478	144	100	37			
Frankfurt	FRA	Germany	Global hub	63.2	+2.6	460	137					33.2
Madrid	MAD	Spain	O&D gateway	68.2	+3.0	431	158	75				
Munich	MUC	Germany	Global hub	43.4		337	129					
Berlin	BER	Germany	O&D gateway	26.1	+2.3	193	135					
Helsinki	HEL	Finland	Regional hub	17.0	+4.1	153	111		33	30	57	
Singapore	SIN	Singapore	Global hub	70.0	+3.4	374	187					29.7
Seoul Incheon	ICN	South Korea	O&D gateway	74.1	+4.1	426	174	99	11	106	70	39.8
Hong Kong	HKG	Hong Kong	Cargo-led	61.0	+15.0	395	155					83.1
Bengaluru	BLR	India	O&D gateway	43.8	+7.6							11.9
MEDITERRANEAN, NORTH AFRICA AND THE LEVANT												
Cairo[d]	CAI	Egypt	O&D gateway	30.9	+6.8	224	138			28	111	
Hurghada[b]	HRG	Egypt	Leisure	11.4	+18.5	77	149					
Sharm El Sheikh[b]	SSH	Egypt	Leisure	9.0	+32.3	60	152					
Istanbul	IST	Türkiye	Global hub	84.5	+5.0	549	154	79		90	94	
Istanbul SAW	SAW	Türkiye	O&D gateway	48.4	+17.0			56				
Antalya	AYT	Türkiye	Leisure	39.0	+2.0			82				
Izmir	ADB	Türkiye	Second city	12.7	+10.0							
Dalaman	DLM	Türkiye	Leisure	5.6	−1.1							
Bodrum	BJV	Türkiye	Leisure	4.4	+1.0							
Athens	ATH	Greece	O&D gateway	34.0	+6.7			72				
Thessaloniki	SKG	Greece	Second city	8.0	+8.2							
Rhodes	RHO	Greece	Leisure	7.0	+2.5							
Larnaka	LCA	Cyprus	Leisure	9.9	+14.5							
Pafos	PFO	Cyprus	Leisure	3.8	+5.6							
Malta	MLA	Malta	Leisure	10.0	+11.6							
Palma de Mallorca	PMI	Spain	Leisure	33.8	+1.5	246	137					
Barcelona	BCN	Spain	O&D gateway	57.5	+4.4	361	159					
Casablanca[i]	CMN	Morocco	Regional hub	11.0						14	79	
Marrakech[i]	RAK	Morocco	Leisure	10.2	+10.1					8	127	
Agadir[i]	AGA	Morocco	Leisure	3.5	+11.1					2.5	140	
Djerba	DJE	Tunisia	Leisure	2.3	+6.5	17	137					
Amman[h]	AMM	Jordan	O&D gateway	9.8	+11.3	81	122			12	82	

Source: Operators and airport groups (Dubai Airports, HIA (Doha), Abu Dhabi Airports, Heathrow, Gatwick, Royal Schiphol Group, Fraport, Aena, Munich Airport, FBB, Finavia, Changi Airport Group via Asian Aviation, IIAC, AAHK, BIAL via Airport World); DHMI; Egypt Ministry of Civil Aviation via Business Today Egypt and Amwal Al Ghad; AIA; Fraport Greece; Hermes Airports; Malta International Airport; ONDA via APA News, Hespress and Finances News; Tunisian Ministry of Transport via La Presse; AACO; Saudi Press Agency via Bernama. Notes: a, Heathrow utilisation is 2025 movements against its 480,000 cap; b, FRACS, Air Transport Data Bulletin (2026), citing operators and ACI; c, Jeddah traffic from the FRACS bulletin (2026), utilisation as stated by GACA; d, Cairo capacity is Terminals 1 to 3, with Terminal 4 under way; h, Amman capacity is the 2007 terminal design; i, ONDA's stated current capacity (2025).

THE HEIRSTONE AIRPORT BENCHMARK 2026, CONTINUED

EXHIBIT 13

The Heirstone Airport Benchmark 2026 (part 2 of 2)

AIRPORT	IATA	COUNTRY	JOB	PAX 2025 (M)	GROWTH (%)	MOVEMENTS (000)	PAX PER MOVEMENT	INTL SHARE (%)	TRANSFER SHARE (%)	CAPACITY (M)	UTILISATION (%)	CARGO T PER 1,000 PAX
SUB-SAHARAN AFRICA												
Johannesburg	JNB	South Africa	Regional hub	19.7	+7.4	209	94					18.9
Addis Ababa	ADD	Ethiopia	Regional hub	13.1	+8.6	183	72			22	60	21.5
Cape Town	CPT	South Africa	O&D gateway	11.1	+7.2	99	113	30				9.1
Nairobi	NBO	Kenya	Cargo-led	9.0	+2.9	121	74			7.5	120	45.3
Lagos	LOS	Nigeria	O&D gateway	7.5	+13.1	96	77					30.4
Abuja	ABV	Nigeria	O&D gateway	5.8	+12.5	94	62			15	39	1.5
Durban	DUR	South Africa	Second city	5.5	+10.4	44	124					2.4
Accra[e]	ACC	Ghana	O&D gateway	3.6	+2.4	47	77	70		5	51	12.4
Dar es Salaam	DAR	Tanzania	O&D gateway	3.0	+4.3	59	51					9.5
Dakar	DSS	Senegal	O&D gateway	2.9	+0.5	25	117			3	98	12.3
Zanzibar	ZNZ	Tanzania	Leisure	2.7	+13.9	68	40					2.0
Entebbe[e]	EBB	Uganda	O&D gateway	2.6	+10.1	41	64			2	112	26.8
Abidjan	ABJ	Côte d'Ivoire	O&D gateway	2.5	+0.9	33	76					13.9
Mombasa	MBA	Kenya	Leisure	1.8	+5.4	28	66					3.3
Harare	HRE	Zimbabwe	O&D gateway	1.7	+8.6	29	58			6	28	7.9
Lomé	LFW	Togo	O&D gateway	1.6	+5.1	18	87					11.0
CENTRAL ASIA AND THE CAUCASUS												
Tashkent	TAS	Uzbekistan	O&D gateway	10.0		80	125	86		11	91	
Samarkand	SKD	Uzbekistan	Second city	1.5		12	125	96		2	76	
Namangan	NMA	Uzbekistan	Second city	0.87		6	151	94				
Urgench	UGC	Uzbekistan	Second city	0.86		7	131					
Fergana	FEG	Uzbekistan	Second city	0.69		6	115					
Bukhara	BHK	Uzbekistan	Second city	0.57		6	99					
Almaty	ALA	Kazakhstan	O&D gateway	11.9	+4.0			46		14	85	
Baku	GYD	Azerbaijan	Regional hub	7.6					13			
Tbilisi and Batumi[f]	TBS	Georgia	O&D gateway	6.6	+16.3							
Yerevan[g]	EVN	Armenia	O&D gateway	5.6	+7.5					4	140	

Source: FRACS, Air Transport Data Bulletin (2026), citing operators and ACI, for Sub-Saharan Africa; KAA master plan (2026); FAAN (2018); Ghana Civil Aviation Authority (2025); UCAA (2025); AIBD and Moodie Davitt Report (2017); CAAZ via Airport Technology; Uzbekistan Airports via Anhor.uz and UzDaily (2026); TAV Airports annual report 2025; Baku airport operator (2026); Corporación America Airports and CivilNet (2026). Notes: e, international traffic against international terminal capacity (Entebbe 2024); f, Tbilisi and Batumi combined (TAV); g, Yerevan traffic rounded, growth for the operator's Armenian airports, capacity is the current terminal's design; Baku transfer share is transit passengers.

READING THE BENCHMARK

Four patterns in the table

The benchmark is a starting point for any airport master plan, concession or investment case in these regions: it shows which airports do a similar job, at what scale, with what fleet and against what capacity.

Global transfer hub · Regional hub · O&D gateway · Leisure gateway · Secondary city · Cargo-led

Jobs differ more than sizes.

Transfer shares run from 75% at Doha to 10.9% at Incheon, and international shares from 29.7% at Cape Town to 95.9% at Samarkand. The job column tells an owner which peers to compare against; the passenger column only sets the scale.

Utilisation needs its basis.

Some capacities are terminal designs, some airport-wide and some first phases. We show the basis in Exhibit 11; leisure airports should also be read against their peak month (Exhibit 10).

Cargo marks a distinct job.

Hong Kong carries 83 tonnes of cargo per 1,000 passengers and Nairobi 45, against 19 at Johannesburg. In East Africa, cargo and logistics land belong in the airport business plan.

Growth sits outside the old hubs.

Sharm El Sheikh grew 32.3%, Hurghada 18.5% and Istanbul's second airport 17%; Abu Dhabi grew 13% after opening its 45 million terminal and Lagos 13.1%, against 0.7% at capacity-capped Heathrow.

CHAPTER 4

04 Good at its job

A transfer hub that held passengers for long dwell times would be failing its airline; a leisure gateway sized for its average day would fail its peak. The same numbers can mean success at one airport and waste at another. We classify each benchmark airport by the job it does, using markers that can be read from published data.

EXHIBIT 14

Six airport jobs, each with its own markers and measures

Heirstone typology of airport jobs; markers are ranges observed in the benchmark

JOB	WHAT IT DOES	BENCHMARK EXAMPLES	MARKERS IN THE DATA	MEASURES THAT MATTER MOST
Global transfer hub	Connects flows between regions on a home airline's banks of flights	Doha, Dubai, Istanbul, Abu Dhabi, Amsterdam, Frankfurt, Singapore; Bishoftu planned	Transfer 35 to 75%; 130 to 210 passengers per movement	Connection time, contact stands in the bank, runway wave capacity, transfer retail
Regional hub	Connects a continent or sub-region while serving its own city	Addis Ababa, Johannesburg, Casablanca, Helsinki, Baku	Transfer 10 to 35%; 70 to 130 per movement	Network breadth, punctuality, flexible stands that serve both narrowbody and widebody
O&D gateway	Serves a capital or primary city and its country	Cairo, Madrid, Berlin, Incheon, Athens, Tashkent, Almaty, Lagos, Accra	Transfer under 15%; international share 30 to 99%	Processing times, kerb and rail access, parking and property, aeronautical yield
Leisure gateway	Brings tourists to a coast, island or heritage site	Antalya, Hurghada, Palma, Marrakech, Larnaka, Rhodes, Zanzibar, Djerba	Peak month 1.6 to 2.2 times the average month; 40 to about 150 per movement	Peak-month capacity, coach forecourts, off-peak yield, retail and food ahead of parking
Secondary city airport	Links a second city to the capital, the diaspora and leisure markets	Samarkand, Izmir, Thessaloniki, Durban, Namangan, Fergana, Bukhara	80 to 150 per movement; over 90% international in parts of Uzbekistan	Cost per passenger of capacity, phasing, airline count, space discipline
Cargo-led gateway	Earns a large part of its business from freight and logistics land	Hong Kong, Nairobi; Lagos and Entebbe in part	30 to 85 tonnes of cargo per 1,000 passengers	Cargo apron and terminal capacity, logistics land, cargo per passenger

Source: Heirstone Airport Benchmark 2026. **Heirstone analysis.** An airport can do more than one job; the benchmark assigns its main one.

Hub labels hide different businesses. Doha is about 75% transfer. Incheon is almost wholly international, with 99.3% of its 74.1 million passengers on international routes, yet only 10.9% transfer: on our classification it is a capital gateway with a strong network, and its benchmarks should come from other gateways. Cairo, with a home carrier, the world's second-largest route by seats to Jeddah and Terminal 4 under way, is a capital gateway building hub capacity.

Uzbekistan's regional airports show why the job matters for planning. Samarkand and Namangan carried 95.9% and 94.5% international passengers in 2025, Tashkent 86.2%, while Almaty, a larger airport, was 45.8%. These are international gateways in their own right, and their terminals, border processing and stand mix should reflect that.

10.9%

transfer share at Incheon in 2025, against about 75% at Doha

5.8M

seats on Cairo to Jeddah in 2025, the world's second-largest route by seats

THE HEIRSTONE AIRPORT TEST

Eight questions any owner can put to an airport, with pass marks from the benchmark and weights by job

EXHIBIT 15

The Heirstone Airport Test

Filled circles mark the questions that matter most for each job; open circles those that matter; blank, those that rarely decide the case

	TEST AND QUESTION	WHAT THE BENCHMARK SHOWS	PASS MARK	GLOBAL HUB	REGIONAL HUB	O&D GATEWAY	LEISURE	SECOND CITY	CARGO-LED
1	**Peak.** Is the terminal sized on a stated design hour taken from the airport's own traffic?	Peak hour 0.024% (Abu Dhabi T-A) to 0.05% (Samarkand) of annual capacity; peak month 1.18x (Istanbul) to 2.23x (Bodrum) the average	Design hour stated and evidenced; above 1.5x, size on the peak month	●	●	○	●	○	○
2	**Headroom.** Will traffic stay within capacity until the next phase opens?	Utilisation 27% to 140%; build times of four to five years	Next phase committed at $1/(1+g)^4$ of capacity: 82% at 5% growth, 68% at 10%	○	●	●	●	●	●
3	**Space.** Does terminal area per passenger sit inside the benchmark band?	9.7 to 16.9 m² per 1,000 annual passengers of capacity	Inside the band, or a stated reason	○	○	●	○	●	○
4	**Stands and gauge.** Are stands sized from the fleet mix and the peak wave?	0.41M to 1.05M passengers per contact stand; 40 to 209 passengers per movement	Stands from design-hour movements by aircraft code; pier-served target written down (95% of passengers at Heathrow)	●	●	○	○	○	●
5	**Runway.** Is runway capacity declared by the hour and used?	Gatwick 55 to 57 movements an hour on one runway; Heathrow 99.6% of its cap	Hourly declared capacity published; second runway justified by peak or resilience	●	○	○			●
6	**Cost.** Is cost per passenger of capacity in range for its scope, with the scope stated?	Expansions US$18.5 to 83; new terminals US$24 to 266; new airports US$125 to 250	Within the scope band; scope published	○	○	●	●	●	○
7	**Commercial.** Is the commercial plan built for how passengers arrive and dwell?	Non-aeronautical 26% (Africa) to 47% (Middle East) of revenue; about US$4 to over US$16 per passenger	Targets set against the regional peer; no parking-led plan at coach-arrival airports	●	○	●	●	○	●
8	**Service and contract.** Are service outcomes published, with money attached?	95% of passengers within 5 to 10 minutes at security; up to 7% of charges at risk	Outcome targets in the contract with 1 to 7% of charges at risk; term matched to capex	●	●	●	○	○	○

Source: Heirstone Airport Benchmark 2026 and the sources for Exhibits 1 to 12. **Heirstone analysis.** g is the expected annual traffic growth rate; four years is a typical build period.

The test does not produce a single score. It shows an owner, lender or grantor which questions decide whether an airport is good at its job. Applied to a leisure gateway, it puts the peak month, headroom, cost and commercial plan first: Antalya passes the peak and headroom questions once its 80 million plan is read against its peak month, and its commercial peers are Mediterranean gateways such as Athens, where non-air activities earned 25.3% of revenue in 2025.

Applied to a secondary city airport such as Samarkand, the test puts cost, space and headroom first. Its 2022 terminal cost US$40 to 65 per passenger of capacity, within the range for new terminals, and its 20.6 m² per 1,000 passengers of capacity, on a trade-press area figure, leaves room inside the building. At 76% utilisation in 2025, headroom is the question to watch.

MEASURING SERVICE

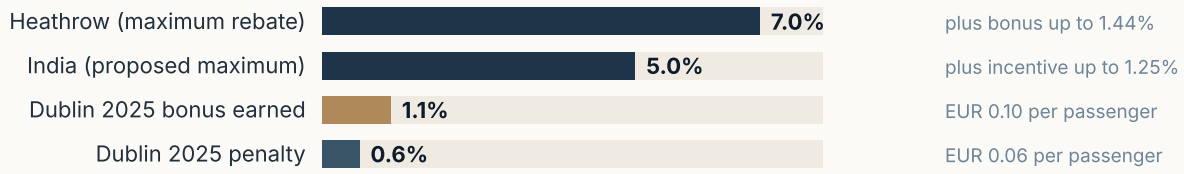
Rankings reward different things; a contract can reward the right ones

EXHIBIT 16

Who measures airport service, and how much money rides on it

Top: the main measures used to compare airport service. Bottom: share of airport charges at risk or earned under service regimes (%)

MEASURE	OWNER	METHOD AND COVERAGE	WHAT IT REWARDS
ASQ awards	ACI World	Survey of about 707,000 passengers at the gate; over 400 airports	Satisfaction relative to region and size band
World Airport Awards	Skytrax	Open online traveller survey covering more than 575 airports	Traveller preference across the journey
Star rating	Skytrax	Audit of up to 800 customer-facing areas	Product and front-line service
On-time performance	Cirium	Departures within 15 minutes, from over 600 data sources	Punctuality only
AirHelp Score	AirHelp	Composite of 279 airports; punctuality 60% of the score	Mostly punctuality
Operator outcomes	Airports (e.g. Dubai)	Queue and baggage times from airport systems	Processing speed
Regulated targets	UK CAA, Irish regulator, AERA (proposed)	Operational and survey targets, reported monthly or quarterly	Meeting targets, with money at risk



Source: ACI World (2026); Skytrax (2026); Cirium (2026); AirHelp (2026); Dubai Airports (2026); Heathrow, Measures, Targets and Incentives; Irish Aviation Authority (2026); AERA consultation (2025). Dublin shares are Heirstone analysis of the bonus and penalty per passenger against the EUR 9.40 price cap.

The yardsticks measure different things and reward different airports. ACI's survey compares satisfaction within a region and size band; Skytrax's awards reflect an open vote; Cirium measures punctuality alone; AirHelp's score is 60% punctuality. Cape Town is the African airport that scores on all four: four ASQ awards for 2025, Skytrax's Best Airport in Africa for 2026, third in AirHelp's 2026 score and sixth among Cirium's small airports. Riyadh was the world's second most punctual large airport in 2025 at 86.81% and Skytrax's most improved. Baku is the only Skytrax five-star airport in Central Asia, the CIS and the Caucasus.

Egypt's recognitions are institutional and verifiable. Cairo holds ACI's Public Health and Safety Readiness Accreditation to 2029, which its operator says no other African airport holds under ACI's updated standards. Cairo and Hurghada won ACI Africa sustainability awards in 2025, with Borg El Arab also honoured, and Le Méridien Cairo Airport was named Skytrax's Best Airport Hotel in Africa in 2026. For owners the practical step is the regulated one: publish outcomes, as Dubai does, and attach money to them, as Heathrow, Dublin and India do.

7%

of airport charges can be rebated at Heathrow when service targets are missed

2029

Cairo's ACI health and safety readiness accreditation runs to 2029

CHAPTER 5

05 Who owns the airport

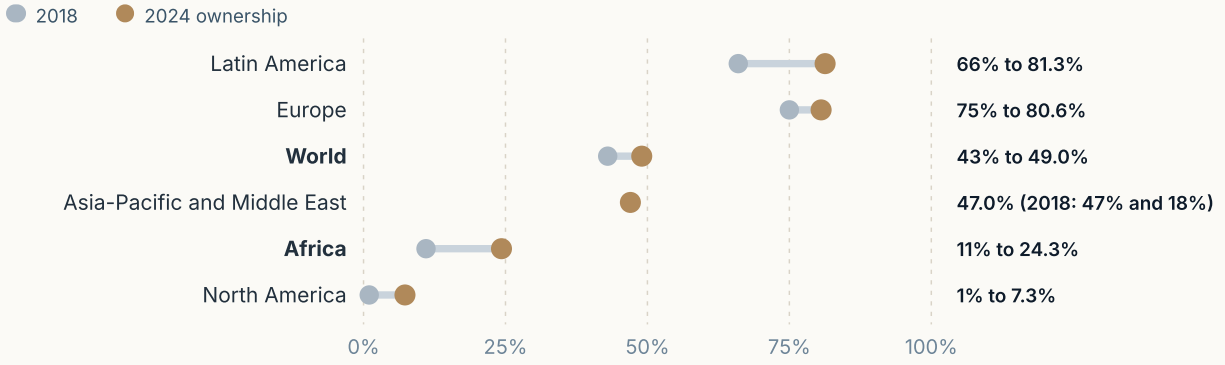
Ownership runs from a government department to a listed company, and most airports sit somewhere between. ACI counted 710 airports with private stakes handling 45% of 2019 traffic; on its 2024 ownership data the share is 49%. Thirty-eight investor-owned airport companies earned US$55.4 billion in 2024, 28.4% of world airport revenue.

The model shapes who carries the cost of the peak. A state company carries it on its balance sheet, a concessionaire prices it into its bid and an airline-owned hub carries it in its network.

EXHIBIT 17

Africa's private share of traffic has more than doubled, to 24.3%

Share of passenger traffic handled by airports with private investment, ACI 2018 data and ACI 2024 ownership data weighted by 2019 traffic (%); 2018 shares were 47% for Asia-Pacific and 18% for the Middle East



Source: Reason Foundation, Annual Privatization Report 2024 and Annual Aviation Infrastructure Report 2026, from ACI data; ACI presentation to the ICAO ESAF workshop (2023).

The spectrum is wide. ACSA, majority-owned by South Africa's government, runs nine airports as a corporatised company; Changi Airport Group is wholly owned by Singapore's government. Management contracts last nine years on ACI's averages, build-operate-transfer concessions 35 and long leases or trade sales 64. Heathrow is fully investor-owned and price-regulated, with a regulatory asset base of £21.3 billion. TAV holds 85% of Almaty as owner, with no concession expiry. Ethiopian Airlines owns and runs its hub, and Qatar Airways holds 60% of Rwanda's new Bugesera airport.

Uzbekistan adds a split model: at the new Tashkent airport a private consortium will build and run the terminal while the state builds and runs the airfield, and the same division applies at Urgench, where Incheon International Airport Corporation will operate the terminals to 2047. Egypt's programme keeps ownership with the state holding company while a private partner upgrades, maintains and operates.

710

airports with private stakes in ACI's 2022 count, handling 45% of 2019 traffic

28.4%

of world airport revenue earned by 38 investor-owned airport companies in 2024

CONCESSION TERMS

The bid variable matters as much as the capital plan

EXHIBIT 18

State take runs from 32% to 54.6% of gross revenue, or fixed fees worth about €7 to €11 a year per passenger

Selected airport concessions, leases and PPPs: model, term, payment to the state and committed investment

AIRPORT	AWARD	MODEL	TERM	PAYMENT TO THE STATE	CAPEX COMMITMENT	HEIRSTONE NOTE
Amman	2007	Rehabilitate, expand, operate	25 years	54.6% of gross revenue	US$675M project; 12M terminal	US$56 per passenger of capacity
Madinah	2011	Build-transfer-operate	25 years	not stated	Over US$1.4bn; 8M to 18M	10 applications, 4 bids; rebalanced 2022
Sofia	2020	Concession	35 years	€281M upfront; 32% of gross revenue, minimum €25M a year	Over €600M	€40 upfront per 2019 passenger
Fraport Greece (14 airports)	2017	Bundled concession	40 years	€1.234bn upfront; €22.9M a year plus 28.5% of EBITDA	About €400M by 2021	€49 upfront per 2016 passenger
Athens	1996	Concession, listed 2024	30 + 20 years to 2046	€1.403bn for the 2019 extension	€1.3bn to 40M by 2032	Valued at €2.46bn at listing
Istanbul	2013	Build-operate lease	25 years	€22.152bn contract price	Phase 1 for 90M	€10.5 a year per 2025 passenger
Antalya	2022	Lease, 2027 to 2051	25 years	€7.25bn, 25% upfront	About €765M; 80M capacity	€7.4 a year per 2025 passenger
Tbilisi	2005	BOT, extended to 2031	to 2031	US$25M upfront; rent 30% of passenger fee from 2027	US$150M; terminal to 57,000 m²	Interim extension, 2026
Yerevan	2001	Concession, extended to 2067	to 2067	not stated	About US$425M over seven years	Extension in return for capex
Almaty	2021	Acquisition (owned)	no expiry	US$422M for 85%	US$200M terminal; €315M programme	Tariff increases to 2030
New Tashkent	2026	Split PPP: private terminal, state airfield	35 years after opening (reported)	not stated	US$2.5bn to 2.6bn stage 1; 20M	US$125 to 130 per passenger of capacity
Hurghada	2025-	PPP tender, IFC lead adviser	to be set	to be set in the request for proposals	Upgrade, maintain, operate	10 applied, 7 prequalified

Source: IFC PPP briefs for Jordan (2009), Saudi Arabia (2012) and Bulgaria (2023) and IFC (2022, 2025); ICAO PPP case study; Growth Fund and Milbank (2024); Fraport Greece (2017); iGA; Fraport and TAV (2021, 2022); TAV annual report 2025; CivilNet (2026); Anhor.uz and Interfax (2025, 2026); Zawya (2026). ICAO records Amman's share as 54.47% then 54.64%. Per-passenger figures are **Heirstone analysis** on stated fees and traffic.

Grantors have three main levers. A share of gross revenue, as at Amman (about 54.6%) and Sofia (32%, with a €25 million floor), shares traffic risk with the state. A share of earnings plus a fixed sum, as with Fraport Greece's €22.9 million a year and 28.5% of EBITDA, rewards cost control. A fixed fee bid up front, as at Istanbul and Antalya, gives the state certainty and leaves the risk with the operator: Istanbul's €22.152 billion over 25 years is about €10.5 a year per 2025 passenger and Antalya's €7.25 billion about €7.4.

Extensions in return for capital are now routine: Athens added 20 years for €1.403 billion in 2019, Yerevan runs to 2067 against about US$425 million of investment, and Tbilisi gained an interim five years in 2026. Madinah's concession was rebalanced in 2022 after traffic fell from 8 million to 2 million. Bundling smaller airports with a stronger one is proven: Fraport Greece runs 14 airports under one concession, and TAV pairs Monastir with Enfidha in Tunisia and Skopje with Ohrid in North Macedonia.

54.6%

of gross revenue to the state at Amman, against 32% at Sofia

14

Greek regional airports bundled in one 40-year concession

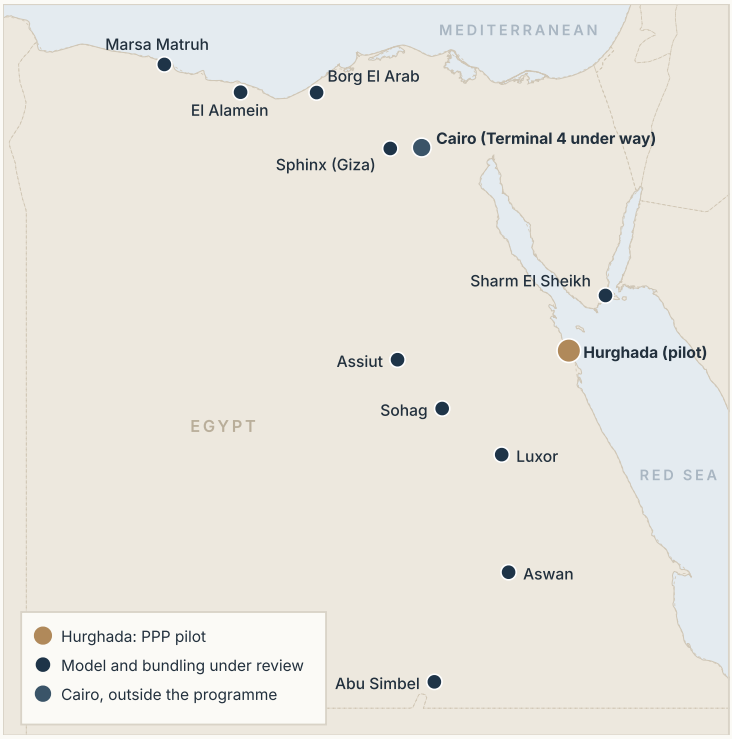
EGYPT'S PROGRAMME

Egypt is tendering eleven airports on a tested model, starting with its busiest leisure gateway

EXHIBIT 19

Eleven airports in Egypt's IFC-advised programme

Airports named in the programme, with Hurghada as the pilot transaction



Source: IFC (2025); Ministry of Civil Aviation via Zawya (2026). Coordinates from GeoNames; Borg El Arab from the GeoNames record used in the benchmark.

Egypt engaged IFC in 2025 to introduce public-private partnerships at 11 airports: Hurghada as the pilot, then Sphinx, Sharm El Sheikh, Borg El Arab, Luxor, Aswan, Sohag, Assiut, Abu Simbel, El Alamein and Marsa Matruh. EHCAAN, the state airports holding company, keeps ownership; the private partner will upgrade, maintain and operate.

Hurghada, which IFC describes as Egypt's second-busiest airport, went to tender in 2025. Ten consortia applied and seven were prequalified in 2026; the request for proposals is the next stage. For the other ten airports IFC will recommend the most suitable form of private participation, examine bundling several airports into one transaction and draft a roadmap.

The design follows structures that have worked elsewhere: the state keeps the assets, as Uzbekistan keeps its airfields; bundles are under study, as used in Greece and Tunisia; and IFC acts as transaction adviser, as it did at Amman, Madinah and Sofia. The programme sits within Egypt's Asset Monetization Program, launched in 2023, and in 2026 the Presidency directed the creation of integrated investment zones around Egyptian airports.

Three choices will set the value of the programme. The first is the bid variable: a revenue share, an EBITDA share with a fixed sum, or a fixed fee. The second is the bundle: pairing smaller Upper Egypt airports with stronger Red Sea and Mediterranean gateways, as Fraport Greece and TAV did, spreads fixed costs and risk. The third is the service regime: published targets with money attached, of the kind Heathrow, Dublin and India use, give the state a lever that outlasts the tender.

Leisure economics matter here. Where passengers arrive mainly by tour transfer, as at many resort airports, parking cannot carry the commercial case; retail, food and beverage, property and charges design will. The new investment zones around airports give the programme a property dimension that most airport concessions lack.

11

airports in Egypt's IFC-advised programme, with state ownership retained

7 of 10

consortia that applied for Hurghada were prequalified (2026)

CHAPTER 6

06

From Casablanca to Almaty

Three regions with three different airport problems: leisure peaks and new capacity around the Mediterranean, mismatched capacity in Sub-Saharan Africa, and fast growth from a small base in Central Asia and the Caucasus.

EXHIBIT 20

The benchmark airports by size and job

Benchmark airports in the Mediterranean, the Gulf, Africa, Central Asia and the Caucasus; circle area shows 2025 passengers, colour shows the Heirstone job classification



Source: Heirstone Airport Benchmark 2026; Riyadh, shown for reference, from the FRACS bulletin (2026); Tbilisi, Batumi and Kutaisi are the first eleven months of 2025. Coordinates from GeoNames; Bishoftu is placed at the town, so its position is approximate. Northern European and Asian reference airports are not shown.

The Gulf is the reference point for all three regions. Dubai handled 95.2 million passengers in 2025 and Dubai World Central is planned for 150 million within a decade; Saudi airports carried almost 141 million (+9.6%), with Jeddah at 107% of capacity according to the Saudi Press Agency; Abu Dhabi grew 13% after its new terminal opened.

The map also shows how concentrated each region is. Outside Johannesburg, no Sub-Saharan airport in the sample handled more than 13.2 million passengers in 2025, and in Central Asia only Almaty and Tashkent neared or passed 10 million. Egypt pairs Africa's busiest airport with a large second tier: the Egyptian Airports Company network carried about 28 million in 2025, led by Hurghada (11.4 million) and Sharm El Sheikh (9.0 million).

THE MEDITERRANEAN AND NORTH AFRICA



Growth moved to the leisure and secondary gateways

Images are illustrative.

58.9M

passengers at Egypt's public airports in 2025, up from about 51.9 million (Heirstone analysis)

+22.3%

growth at Egypt's airports other than Cairo in 2025, to about 28 million

36.3M

passengers at Morocco's airports in 2025, a record (+11%)

On the national totals we could verify, Egypt runs the largest airport system in Mediterranean Africa. Its public airports handled about 58.9 million passengers in 2025: 30.94 million at Cairo, Africa's busiest airport, and about 28 million across the Egyptian Airports Company network. That is about 1.6 times Morocco's record 36.3 million and well above Tunisia's 12.5 million in 2024, the two other North African national totals we could verify. Egypt received about 19 million tourists in 2025, a record and 21% more than in 2024.

The growth came from the gateways. Egypt's airports outside Cairo grew 22.3% in 2025 against 6.8% at Cairo; on FRACS figures Sharm El Sheikh grew 32.3% and Hurghada 18.5%, which supports the choice of Hurghada as the first private partnership. Across the northern shore, Istanbul's second airport grew 17%, Larnaka 14.5% and Malta about 12%, while Istanbul, Barcelona and Antalya grew 2 to 5%.

Türkiye's airports carried 246.8 million passengers in 2025 and Greece's two largest operators 71.1 million between them. The northern shore has the scale. On the southern shore, Cairo and Morocco's network both plan to more than double their capacity.

The leisure airports on both shores share one problem: the peak. Antalya, Dalaman and Bodrum move most of their year in four summer months, and Cyprus about half. That makes peak-month capacity, coach and transfer forecourts and off-season demand the central planning questions, and it makes monthly traffic data the most useful single thing a grantor can publish.

On the eastern and southern shores, Amman carried 9.79 million passengers in 2025, up 11.3%, and its operator added a 4.8 MW solar farm supplying nearly a quarter of the airport's operational energy. Djerba carried over 2.3 million, up 6.5%, with a modernisation budget of TND 77.6 million to 2030.

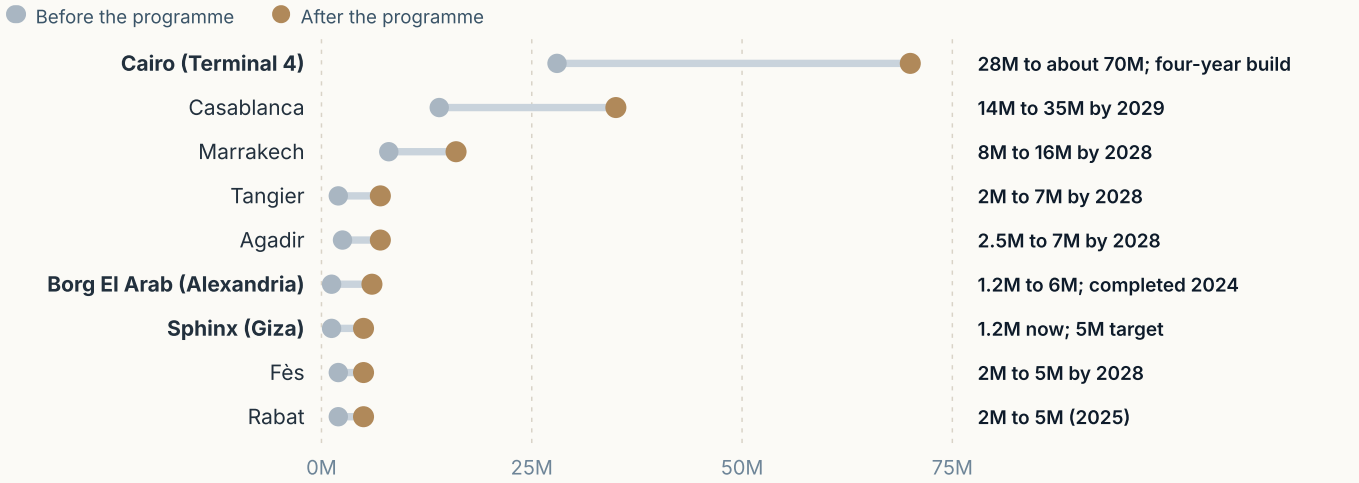
THE MEDITERRANEAN AND NORTH AFRICA: CAPACITY

Cairo's Terminal 4 is the largest single-airport capacity step on the southern shore

EXHIBIT 21

Egypt and Morocco are adding the most capacity on the southern shore

Stated airport capacity before and after announced or completed programmes (million passengers a year)



Source: Egypt Ministry of Civil Aviation via Ahram Online (2025, 2026), Egypt Today (2025, 2026) and State Information Service (2025); Cairo Governorate (2024) and Orascom Construction for Borg El Arab; Egyptian Streets and Enterprise (2025) for Sphinx; ONDA via Finances News Hebdo (2025) and Maroc.ma (2025). Morocco's current capacities as stated by ONDA.

Cairo carried 30.94 million passengers in 2025 on 224,303 flights, ahead of the 28 million design capacity of Terminals 1 to 3: demand running ahead of the programme now under way. Terminal 4 was scaled from at least 30 million passengers in 2025 statements to about 40 million in 2026, which takes the airport to roughly 70 million. The Minister of Civil Aviation puts its cost at US$4.0 to 4.5 billion over about four years, with phased operation possible after two and a half to three years; the State Information Service gave a minimum of US$3.5 billion, self-financed by EHCAAN with national entities. A cargo city alongside starts at 1 million tonnes a year, expandable to 2 million.

Cairo is also gaining a second gateway. Sphinx International, at kilometre 45 of the Cairo to Alexandria desert road, serves the Grand Egyptian Museum and the pyramids. Its capacity rose from 0.9 to 1.2 million in 2025, with a target of 5 million within five years, and its 2025 traffic passed 1 million well before the year ended, up 21%, according to Enterprise and Egyptian Streets. Like Istanbul, whose second airport carried 48.4 million passengers in 2025, Cairo can then focus its main airport on hub and long-haul traffic. Borg El Arab's 40,000 m² Terminal 2, completed in 2024, takes Alexandria to 6 million.

Morocco plans to lift its network capacity from 34 million in 2024 to 80 million by 2030, including a MAD 15 billion terminal at Casablanca adding 20 million passengers for 2029. Athens is spending about €1.3 billion to reach 40 million by 2032.

70M

Cairo's approximate capacity once Terminal 4, of about 40 million, opens

1M t

initial capacity of the cargo city planned with Terminal 4, expandable to 2 million

+14.3%

growth of Egypt to Dubai traffic in 2025, to 1.8 million passengers

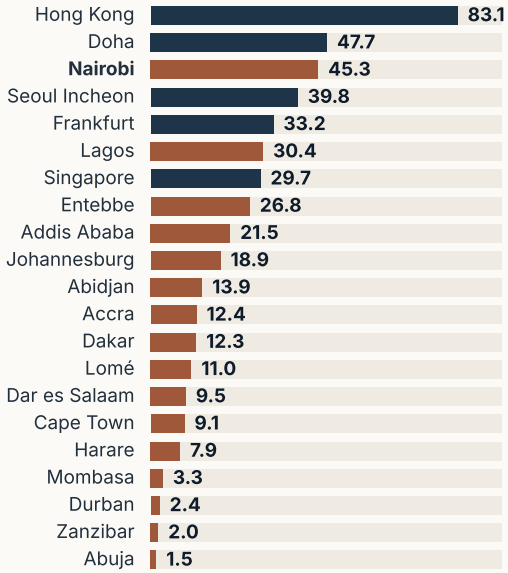
SUB-SAHARAN AFRICA

Sub-Saharan capacity and demand often sit apart, and Nairobi leads on cargo

EXHIBIT 22

Nairobi carries more cargo per passenger than Frankfurt or Singapore

Cargo tonnes per 1,000 passengers, 2025



Source: FRACS bulletin (2026) for African airports; HIA (Doha), Fraport, Changi Airport Group via Asian Aviation, Incheon via Korea JoongAng Daily, AAHK (2026). **Heirstone analysis.**

ACI expected Africa's airports to handle 273 million passengers in 2025, about 2.7% of world traffic. The system is concentrated: Johannesburg, Cape Town and Durban alone carried about 36 million, roughly 13% of the continent's total. Growth in 2025 was strongest at Lagos (+13.1%), Abuja (+12.5%), Zanzibar (+13.9%), Durban (+10.4%) and Entebbe (+10.1%); Dakar and Abidjan grew by under 1%.

Capacity and demand are mismatched. Nairobi ran at about 120% of its stated terminal capacity and Entebbe's international traffic at about 112%, while Harare used 28% of its expanded capacity and Abuja 39% of its new terminal.

Nairobi handled 407,216 tonnes of cargo in 2025, more than Johannesburg's 372,767 tonnes with under half its passengers: 45 tonnes per 1,000 passengers against 19. For East Africa, cargo aprons, cold chain and logistics land are part of the core airport business.

Ethiopia is building the largest project in the Sub-Saharan sample. Bishoftu's first phase, under construction since 2026 for an opening around 2030, is planned for 60 million passengers, 4.6 times Addis Ababa's 13.1 million in 2025; at Addis Ababa's 2025 growth rate of 8.6% that volume would arrive only in the early 2040s, so the ramp-up depends on transfer growth and on phasing. The AfDB plans US$500 million and aims to mobilise about US$8 billion of a US$10 billion cost; the architect cites US$12.5 billion. Rwanda's Bugesera, 60% owned by Qatar Airways, is planned for 8 million passengers rising to over 14 million.

Financing follows three routes. China Exim Bank financed many second-tier terminals, including Nigeria's four new terminals (a US$500 million loan with US$100 million of federal funding) and Harare's expansion (US$153 million). Accra's Terminal 3 drew on a US$400 million loan from commercial banks and the AfDB, according to IOL. Airline equity backs Bugesera. Charges are a lever: AFRAA found African passengers paid an average US$68 in taxes, fees and charges per international departure in 2024, against US$34 in the Middle East. Its 2025 airport-charges index puts Johannesburg, Addis Ababa and Cairo below the African average. Island gateways deserve hub-grade attention: Zanzibar carries 89% of Dar es Salaam's traffic and Mauritius receives 98.3% of its tourists by air.

273M

passengers expected at Africa's airports in 2025 (ACI), about 2.7% of world traffic

4.6x

Bishoftu's first phase against Addis Ababa's 2025 traffic

51

African airports hold Airport Carbon Accreditation, against 294 in Europe

CENTRAL ASIA AND THE CAUCASUS

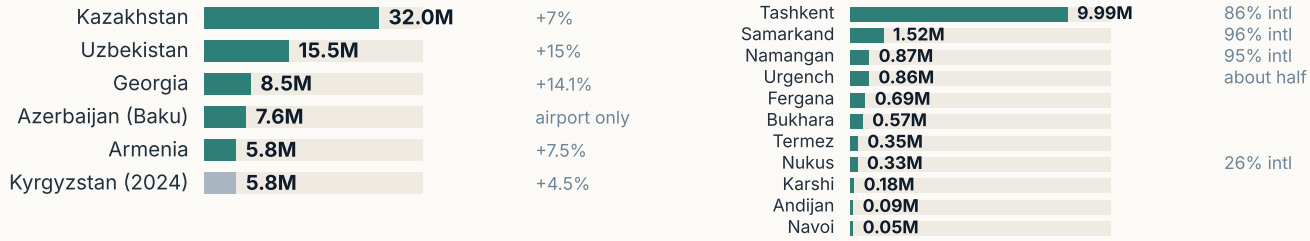


Fast growth from a small base, and almost every ownership model in one region

EXHIBIT 23

Uzbekistan grew 15% in 2025, and its regional airports are international gateways

Left: airport passengers by country, 2025 unless stated, with growth. Right: Uzbekistan's eleven airports, 2025, with international share where published



Source: Uzbekistan Airports via Kun.uz, Anhor.uz and UzDaily (2026); Prime Minister of Kazakhstan (2026); Geostat via GBC (2026); Baku airport operator (2026); Corporación America Airports (2026); Airports of Kyrgyzstan via Economist.kg (2025). Images are illustrative.

Traffic is concentrated in single gateways. Tashkent carried 64.4% of Uzbekistan's passengers in 2025 and officials describe it as at full capacity at about 11 million a year; the first stage of a new airport for 20 million, with two 4 km runways, a 208,000 m^2 terminal and 98 aircraft stands, is due at the end of 2030 at US$2.5 to 2.6 billion. Almaty carried 37.3% of Kazakhstan's traffic and reached 85% of its new 14 million capacity a year after opening. Yerevan carried 5.6 million through a terminal designed for 4 million.

The region shows nearly every model in Chapter 5. TAV owns 85% of Almaty, bought in 2021 for US$422 million, with tariff increases tied to investment to 2030, and runs Tbilisi and Batumi under build-operate-transfer terms. Corporación América holds Yerevan to 2067. Uzbekistan gives private partners the terminals and keeps the airfields, and its draft 2030 strategy aims to pass at least one airport a year to a private operator. Kyrgyzstan's airport company has spent KGS 8.19 billion of its own funds since 2023, about US$94 million, including work at Bishkek and Osh.

Quality stands out: Baku is the region's only Skytrax five-star airport, named Best Airport in Central Asia and the CIS for a ninth successive year, and Tbilisi won five ACI service awards for 2025.

64.4%

of Uzbekistan's passengers used Tashkent in 2025 (Heirstone analysis)

US$40 to 65

Samarkand's 2022 terminal cost per passenger of capacity, against US$125 to 130 for new Tashkent

CHAPTER 7

07 What to decide, and when

Most of an airport's performance is fixed before the first passenger arrives: in the design hour chosen, the job assumed, the contract signed and the land reserved.

FOR

Governments and airport owners

1 Publish the design hour, the design day and monthly traffic for every airport, as DHMI, Aena and Hermes do.

2 Commit the next phase while headroom remains: at 10% growth and a four-year build, at about 68% of capacity.

3 Treat rail and road access as part of the airport at sites far from the city, as Astana did with its 22.4 km airport light rail.

FOR

Concession grantors and DFIs

1 Choose the bid variable deliberately: revenue share (Amman 54.6%, Sofia 32%), EBITDA share plus a fixed sum (Fraport Greece) or a fixed fee (Istanbul, Antalya).

2 Bundle smaller airports with a stronger one, as in Greece, Tunisia and North Macedonia, and plan a mid-life rebalancing route from day one.

3 Write published service outcomes into the contract with 1 to 7% of charges at risk.

FOR

Investors and operators

1 Underwrite leisure airports on their peak month: Antalya's is 1.83 times its average month.

2 Set commercial targets against the regional peer: about US$4 per passenger in Africa and over US$16 in the Middle East, against a world average of US$7.57.

3 Value the tariff path as well as the traffic: Almaty's concession links tariff increases to investment through 2030.

FOR

Developers of airport-city land

1 Property is 15.8% of airport commercial revenue worldwide; Groupe ADP earned €358 million from real estate in 2025.

2 Match land release to the airport's job: logistics land at cargo-led airports such as Nairobi, hotels and offices at capital gateways.

3 Reserve land for the runway and terminal phases first, as Bishoftu's four-runway plan and the 57 km² masterplan for Riyadh's new airport do.

FOR

Tourism authorities

1 Plan the season with the airport: peak-month capacity and off-peak demand are one problem.

2 Watch the cost of arrival: African passengers paid an average US$68 in taxes and charges per international departure in 2024.

3 Use dedicated gateways for major attractions, as Egypt is doing with Sphinx for Giza and the Grand Egyptian Museum.

BY STAGE

Twenty decisions across the life of an airport, each with a benchmark

STAGE 1 Plan

Before the master plan is fixed

Name the job
Classify the airport by job and pick peers with the same markers (Exhibit 14).

State the design hour
From the airport's own data; the FAA ratio is 0.035% of annual traffic at 30 million and above.

Size the peak month
For leisure airports, size on the busiest month multiplied by twelve.

Set the headroom rule
Commit at 82% of capacity at 5% growth, 68% at 10%.

Reserve the land
Protect land for the last phase, as Bishoftu's four-runway and Dubai World Central's five-runway plans do.

STAGE 2 Procure

Before the tender is issued

Pick the bid variable
Revenue share, EBITDA share or fixed fee, with traffic risk placed deliberately.

Design the bundle
Pair weaker airports with a stronger anchor; 14 airports in Greece, two in Tunisia.

Attach money to service
95% within 5 to 10 minutes at security; 1 to 7% of charges at risk.

Match term to capex
ACI averages: 9 years for management contracts, 35 for BOT, 64 for leases.

Plan the rebalancing
Write in the route to a mid-life extension or rebalancing; Athens, Yerevan and Madinah all used one.

STAGE 3 Design

Before the concept is frozen

Hold the space band
9.7 to 16.9 m² per 1,000 annual passengers of capacity, or a stated reason.

Stands from the fleet
0.4 to 1.05 million passengers per contact stand, set by aircraft size.

Walking distance
Moving walkways beyond 300 metres; publish walk times, as Doha does.

Accessibility
UK assistance requests rose 115.8% between 2015 and 2025 against 23.9% traffic growth (CAA data).

Grow along a pier
Choose a shape that can extend by pier or satellite, as Doha, Munich and Incheon did.

STAGE 4 Operate

From the first year

Publish outcomes
Queue and baggage times, as Dubai does every year.

Publish monthly traffic
By airport, with movements and cargo, for every airport in the system.

Track commercial yield
Per passenger, on one definition: airport revenue, with retailer sales reported separately.

Certify carbon
Africa has 51 airports in Airport Carbon Accreditation; new terminals can certify from day one.

Declare hourly capacity
Publish terminal limits in passengers an hour as well as runway slots, as Gatwick's coordinator does.

How Heirstone works on airports and airport land

We help governments, airport owners, investors and developers decide what to build and what it is worth: demand and market studies by airport job and season, benchmarking of the kind in this report, highest-and-best-use and land strategy for airport cities and logistics zones, feasibility and business models (DCF, IRR, NPV and Monte Carlo testing), concession and PPP structuring support, and destination studies that link airports to tourism.

Get in touch

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NOTES, SOURCES AND DEFINITIONS

Notes

Definitions

Passengers: arriving plus departing passengers as published by the operator or authority, 2025 unless stated. **Movements:** aircraft take-offs and landings as published. **Capacity:** stated or design capacity in annual passengers; definitions differ between terminal, airport and phase. **Utilisation:** passengers divided by stated capacity. **Design hour:** the hour a facility is sized for; **typical peak hour (FAA):** the peak hour of the average day of the peak month; **30th busiest hour:** the British standard busy rate. **Level of Service:** IATA's rating of space and waiting time by processor. **Contact stand:** an aircraft stand served by a boarding bridge or pier. **O&D:** origin and destination traffic. **Non-aeronautical (commercial) revenue:** retail, food and beverage, parking, property and other income other than charges on airlines and passengers. **BOT:** build-operate-transfer.

Method

The Heirstone Airport Benchmark 2026 combines operator, authority, regulator and ACI data for 65 airports. Where we derive a figure (passengers per movement, utilisation, cargo per 1,000 passengers, peak-month equivalents, space and cost per passenger of capacity, per-passenger concession fees, headroom points) it is labelled Heirstone analysis and uses only the published inputs shown. Monthly figures for Turkish airports were derived by differencing DHMI's cumulative files. Costs are as published and not adjusted for inflation; scopes differ and are noted. Where sources conflict we show a range or the more conservative figure: Cairo's 2025 traffic is the Ministry's 30.94 million; the FRACS Air Transport Data Bulletin (2026), citing operators and ACI, gives 31.1 million, the highest of any African airport in its table, which is the basis for calling Cairo Africa's busiest airport. Hurghada, Sharm El Sheikh, Jeddah and Riyadh traffic is from the same bulletin. Currency conversions use ECB reference rates in 2026. Single-source figures graded lower in our register are attributed in the text. The airport jobs and the Heirstone Airport Test are Heirstone frameworks.

Use of this report

This report is published for general information. It is not investment, legal, engineering or valuation advice and should not be relied on as such. Figures are as published by the sources named and current in 2026. © Heirstone Consulting L.L.C-FZ, 2026. Images are illustrative.

HOW AN AIRPORT WORKS

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BENCHMARK AND COMPARISONS

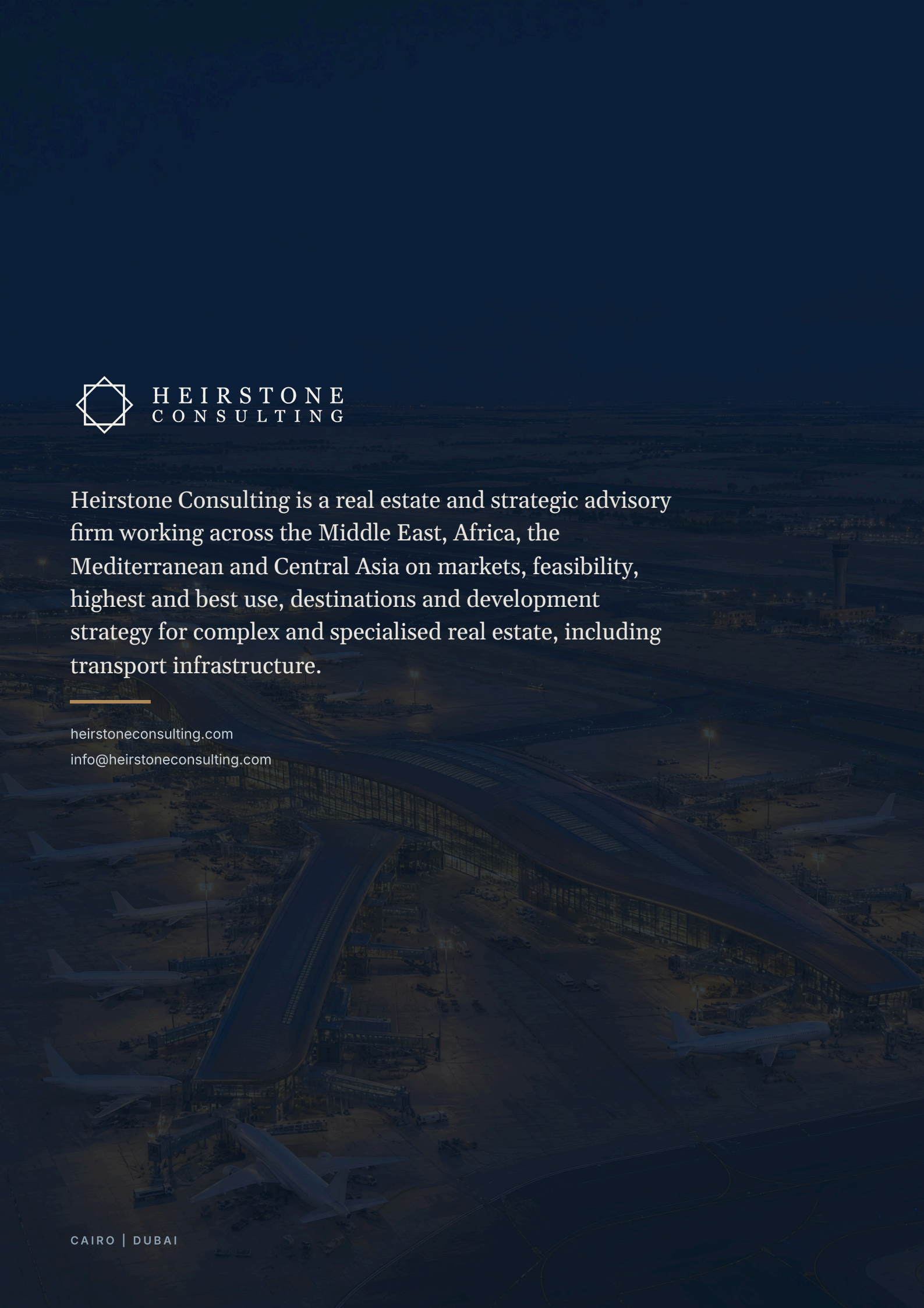
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